SCHEME OF VALUATION - QP Code: 25901045
MG1SDCIIM102 - Mathematics for Instrumentation

Maximum marks: 70 Marks **Time :2Hours**

PART A – Very Short Answers (Any 5 × 2 = 10 marks)

1.A **vector quantity** is a physical quantity that has both magnitude and direction
Example: **Force**, velocity, acceleration, displacement, and momentum are all vector quantities.
Definition-1 marks, Example -1 mark

2. The magnitude of the cross product of two vectors, represents the **area of the parallelogram** that has vectors $\vec{a}$ and $\vec{b}$ as adjacent sides.
Definition-1 marks, Example -1 mark

3. The adjoint of a matrix A, denoted as adj(A), is the **transpose of its cofactor matrix**.adj$(A) = C^T = \begin{matrix} 9 & 3 & 6 \\ 8 & 4 & 0 \\ 7 & 1 & 5 \end{matrix}$

4. A **column matrix** is a matrix that consists of only one column and one or more rows.
Definition-1 marks, Example -1 mark

5. For correct answer-2 marks -$y = \frac{x^4}{4} + C$

6. For any correct definition- 2 marks $\frac{dy}{dx} = \lim_{h\to 0} \frac{f(x+h)-f(x)}{h}$

7. $A + 0 = A$ – 1 mark ;$A \cdot 0 = 0$ – 1 mark

8. For steps -1 mark. For correct answer -1

PART B - Short Answers (Any 5 × 6 = 30 marks)

9. Dot product $\vec{a} \cdot \vec{b} = -9$ -2 mark; Cross Product $\vec{a} \times \vec{b} = -8i - 22j - 13k$ -2 marks; for comments not parallel and not perpendicular- 2marks.

10. For correct steps -4marks. For correct answer $\vec{a} + \vec{b} - \vec{c} = -\hat{\imath} + 2\hat{\jmath}$ -2 marks

11. For finding $B = \frac{1}{2}[A + A^T]$ -4 marks. For verifying $B^T = B$ - 2 marks

12. Correct definition of idempotent matrix -2 marks ; $A^2 = \begin{matrix} 3 & 16 & 6 \\ 2 & 17 & 6 \\ 6 & 12 & 7 \end{matrix}$ -2 marks

13. For formula $\int x^n dx = \frac{x^{n+1}}{n+1} + C$ -2 marks; for steps -2 marks;
For correct answer $\int (x^{-5} + 2x^3 - 5x^2 + 2x)dx = \frac{-x^{-4}}{4} + \frac{x^4}{2} - \frac{5x^3}{3} + \frac{2x^2}{2} + \mathbf{C}$ -2marks

14 .For finding $\frac{dy}{dx} = \frac{d}{dx}(3x^2 - 5x + 1) = 6x - 5$ – 4 marks. For writing $6x - 5 = 0$ – 2marks
Correct answer: The graph has a horizontal tangent at the point $(\frac{5}{6}, -\frac{13}{12})$

15. For table F -2marks; For table G -2marks; For table F+G, F.G -2marks

16. For steps in sum of product expansion for the function $F(x, y, z) = \bar{x} + y + \bar{z}$ -4 marks.
For correct answer $F(x, y, z) = \mathbf{\bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + \bar{x}y\bar{z} + \bar{x}yz + x\bar{y}\bar{z} + xy\bar{z} + xyz}$ **-2 marks**

PART C - Essay Answers (Any 2 × 15 = 30 marks)

17. For writing the formula Area $= \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}|$ -2 marks;
For evaluating$\overrightarrow{AB} \times \overrightarrow{AC} = 0\hat{\imath} - 2\hat{\jmath} + 2\hat{k}$-4 marks;
For finding $|\overrightarrow{AB} \times \overrightarrow{AC}| = \sqrt{0^2 + (-2)^2 + 2^2} = \sqrt{0 + 4 + 4} = \sqrt{8} = 2\sqrt{2}$ -1 mark

For correct answer: Area $= \frac{1}{2}|\overrightarrow{AB} \times \overrightarrow{AC}| = \frac{1}{2}(2\sqrt{2}) = \sqrt{2}$square units -1 marks

For finding $\overrightarrow{AB} \cdot \overrightarrow{AC} = (1)(0) + (-1)(2) + (-1)(2) = 0 - 2 - 2 = -4 \neq 0$ -2 marks

For finding $\overrightarrow{AB} \cdot \overrightarrow{BC} = (1)(-1) + (-1)(3) + (-1)(3) = -1 - 3 - 3 = -7 \neq 0$ -2 marks

For finding $\overrightarrow{AC} \cdot \overrightarrow{BC} = (0)(-1) + (2)(3) + (2)(3) = 0 + 6 + 6 = 12 \neq 0$-2 marks

For the comment not right-angled triangle -1 mark

18. i.) Cramer's Rule $x = \frac{D_x}{D}, y = \frac{D_y}{D}$ - 1 marks.

2 marks for $D_x = \begin{vmatrix} 1 & -3 \\ 3 & 1 \end{vmatrix} = (1)(1) - (-3)(3) = 1 + 9 = 10$

2 marks for $D_y = \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} = (4)(3) - (1)(2) = 12 - 2 = 10$

2 marks for $D_y = \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} = (4)(3) - (1)(2) = 12 - 2 = 10$

For correct answer-x=1 and y=1 - 1 mark

ii) Matrix Inversion Method.

2 marks for writing $X = A^{-1}B$

4 marks for finding $A^{-1} = \frac{1}{10}\begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}$

2 marks for$X = \begin{bmatrix} x \\ y \end{bmatrix} = A^{-1}B = \frac{1}{10}\begin{bmatrix} 1 & 3 \\ -2 & 4 \end{bmatrix}\begin{bmatrix} 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

19. i) Product Rule -2 marks

Steps -3marks

Correct answer $\frac{\mathbf{d(uv)}}{\mathbf{dx}} = \mathbf{2(x+3)(x+4x^2)(12x^2+26x+3)}$ -2 marks

ii) Quotient rule states – 2 marks

Steps -4 marks

Correct answer $\frac{\mathbf{d(u/v)}}{\mathbf{dx}} = \frac{\mathbf{2(x+3)(-4x^2-24x-3)}}{\mathbf{(x+4x^2)^3}}$ -2 marks

20. Logic circuits (15 marks)

(i) AND gate explanation (3 marks)

(ii) Construct circuit for given output (6 marks)

(iii) Only OR + Inverter for 3 variables (6 marks)